IT1244

Artificial Intelligence: Technology & Impact

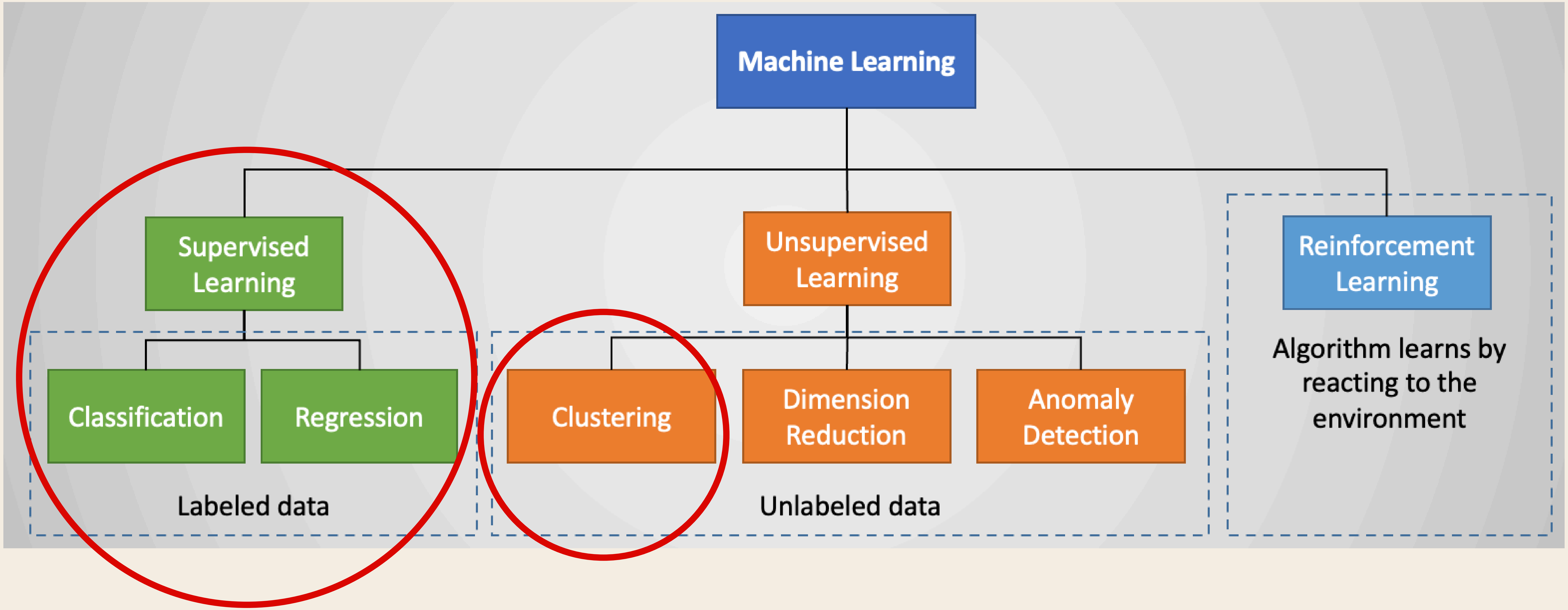
Week 6 Tutorial 4

TODAY'S AGENDA

1. Recap of concepts
2. Discussion of tutorial questions
3. Pollev Questions

RECAP

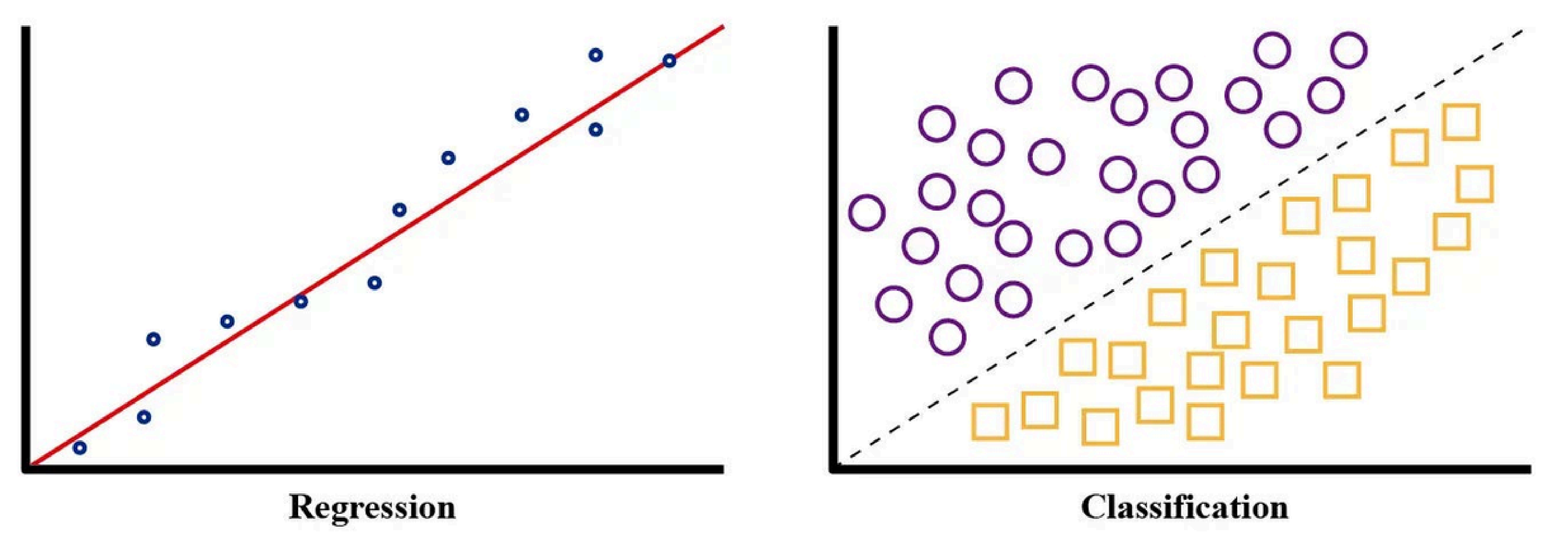
TYPES OF MACHINE LEARNING ALGO



MAIN FOCUS OF IT1244

SUPERVISED LEARNING

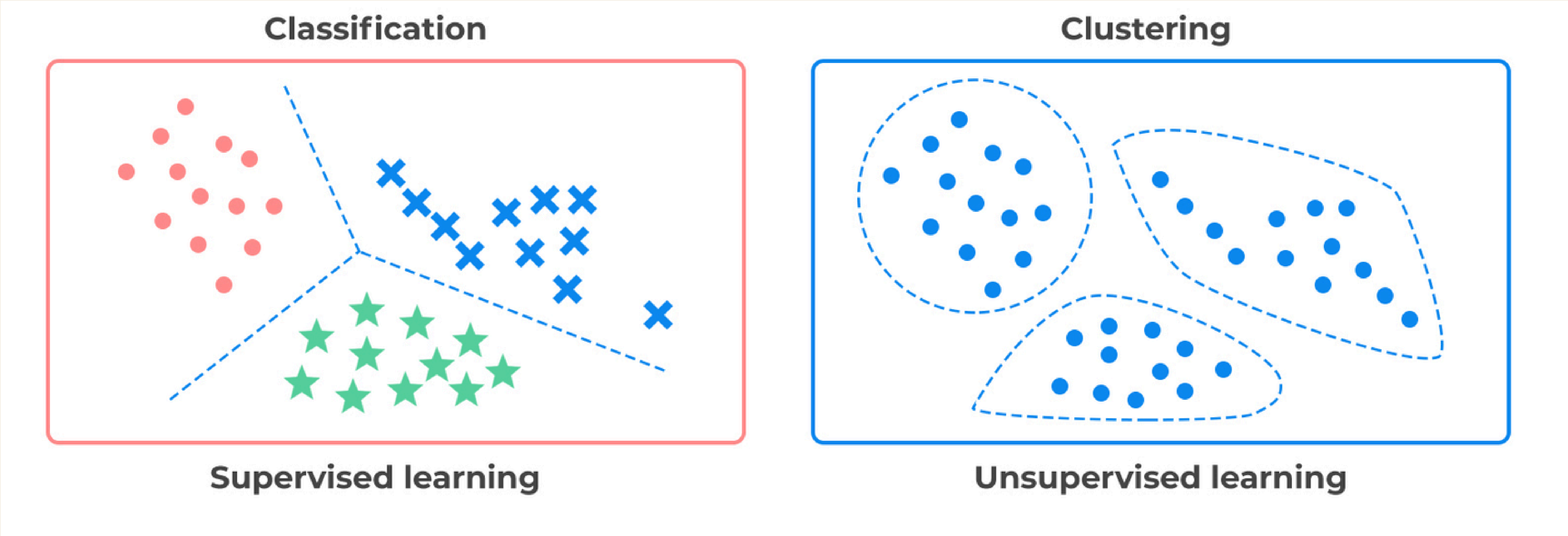
- Goal: Learns from labeled data to make predictions
- Training data includes correct answers (target labels)
- Two main tasks: Regression (continuous values) and Classification (categories)



- Common algorithms: KNN, Linear Regression, Logistic Regression, Decision Trees, Random Forest, Neural Networks

UNSUPERVISED LEARNING

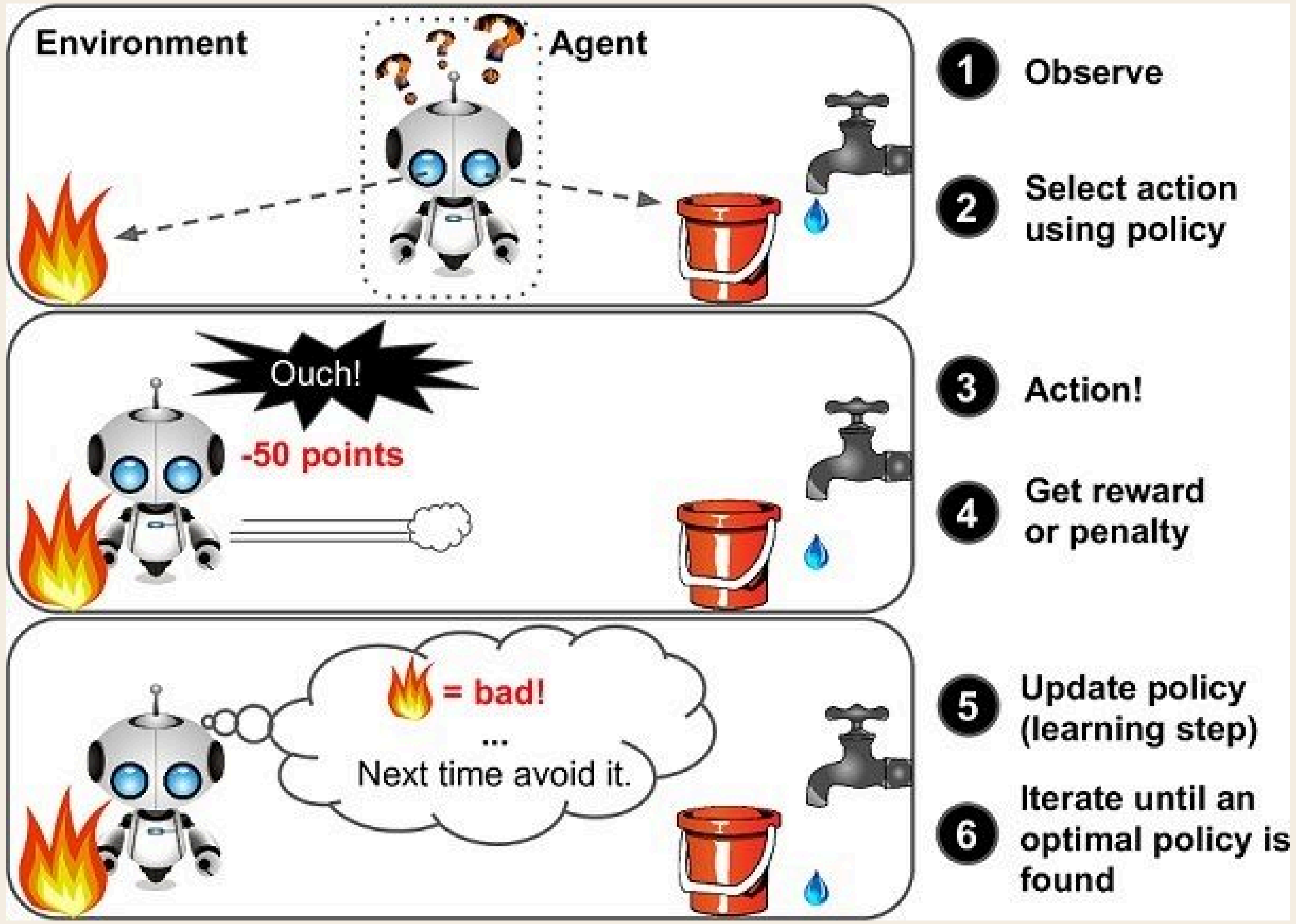
- Learns from unlabeled data (no target labels provided)
- Goal: Discover hidden patterns, structures, or relationships in data



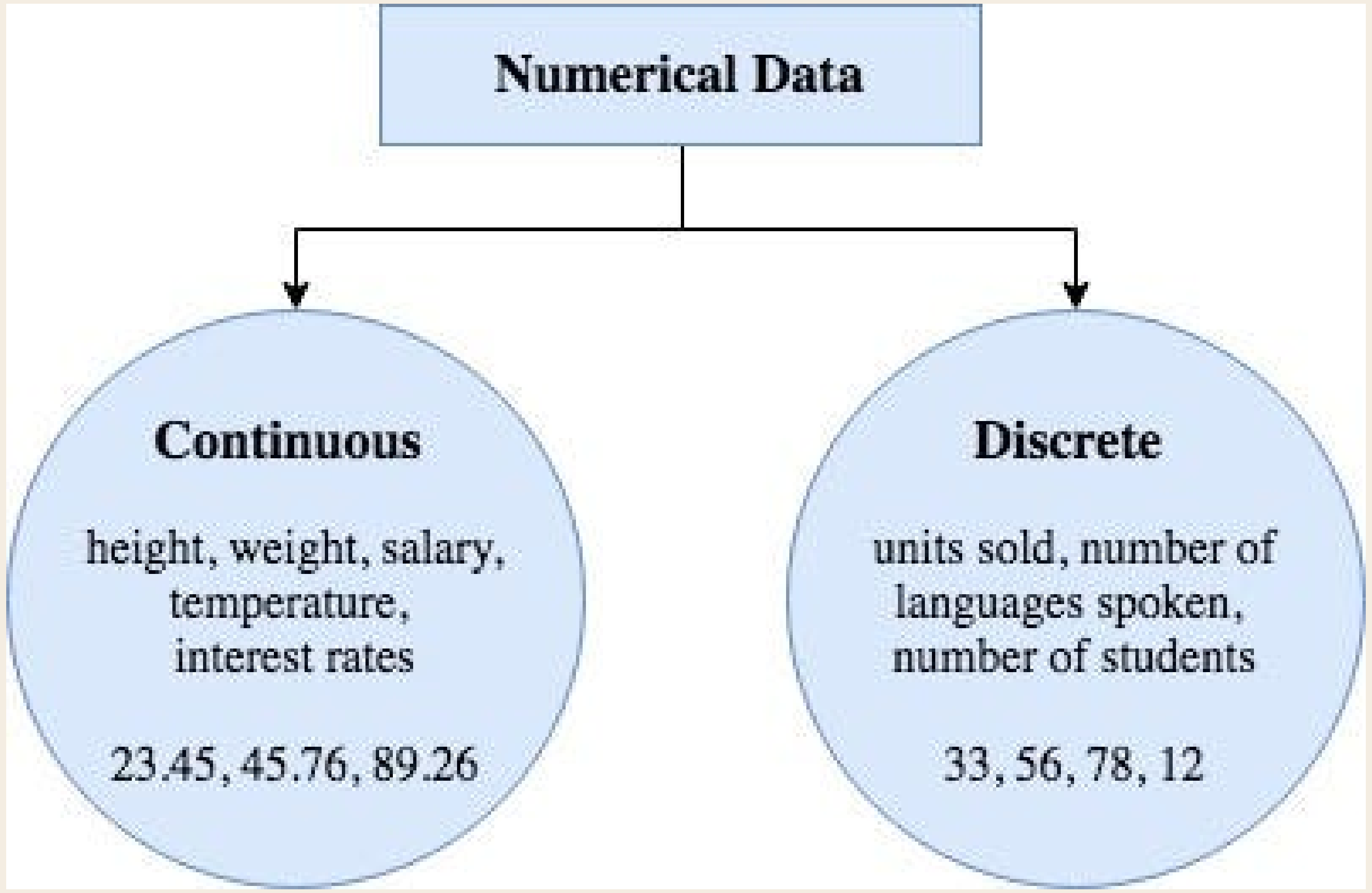
- Common algorithms: K-Means, Hierarchical Clustering, DBSCAN, PCA, t-SNE
- Useful when labeled data is unavailable or too costly (which is very common irl!)

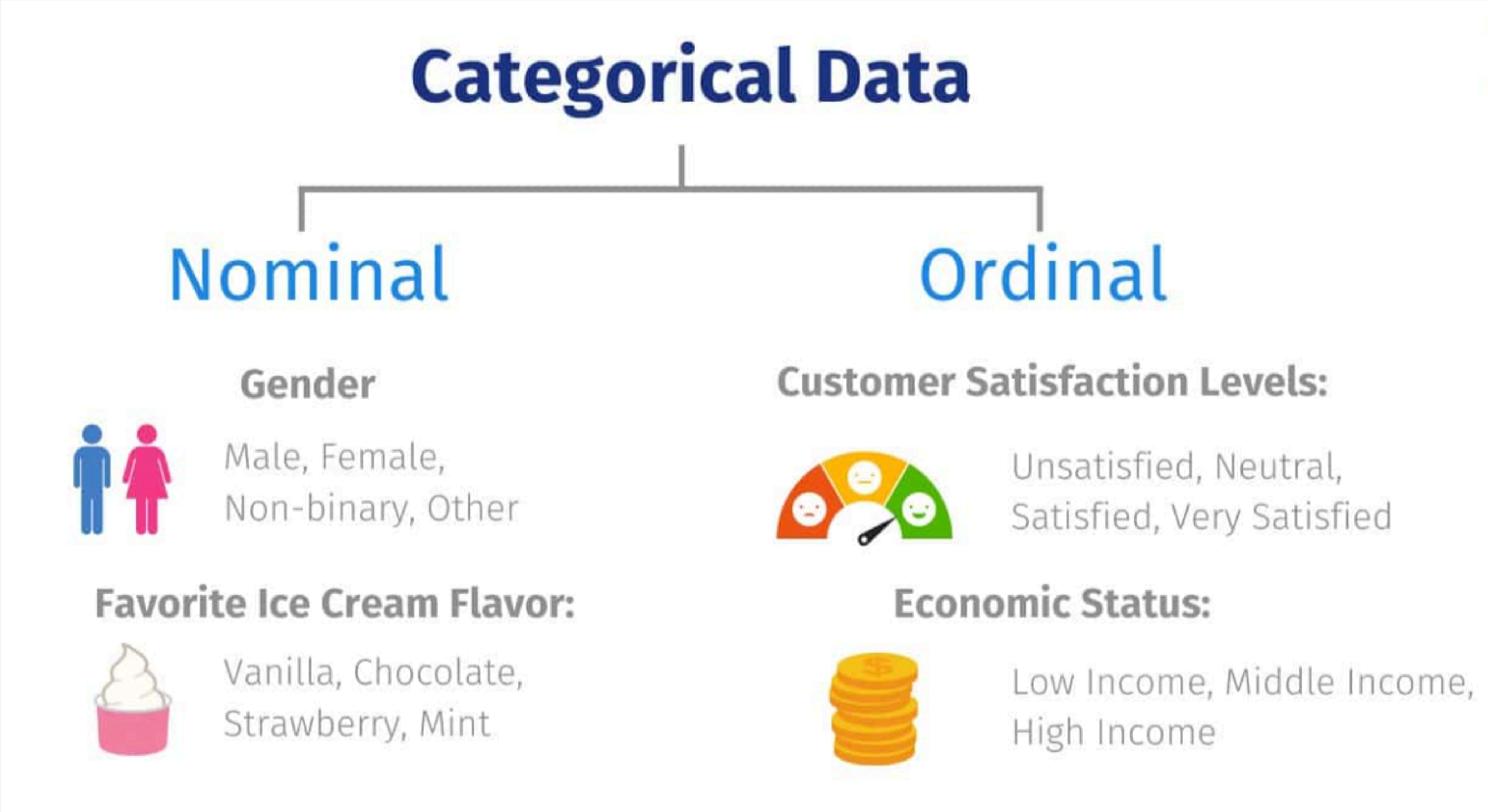
REINFORCEMENT LEARNING

- Learns through trial and error by interacting with an environment
- Goal: Learn actions to maximize cumulative reward / minimize penalties
- Agent receives feedback as rewards or penalties for actions taken

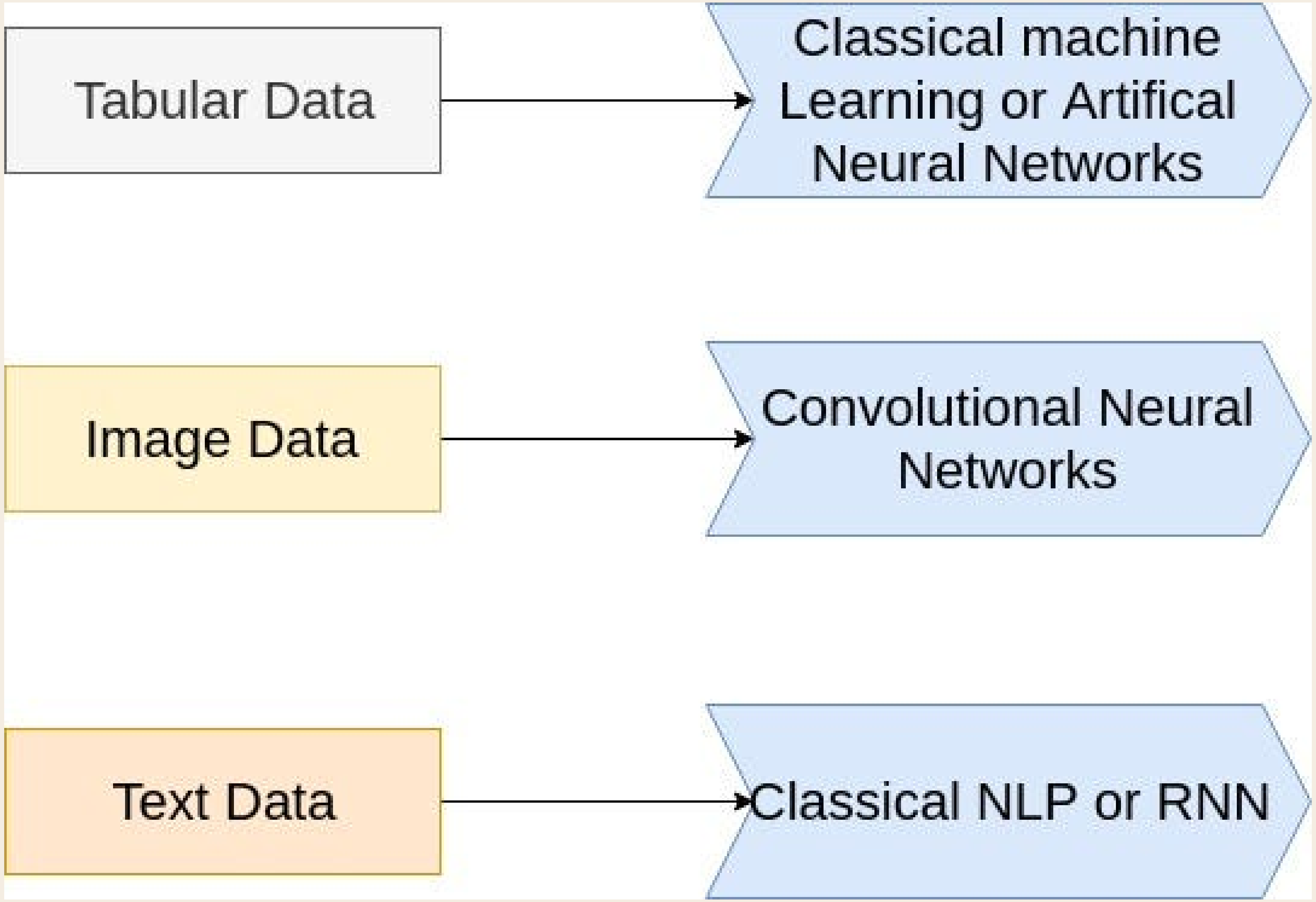


TYPES OF DATA





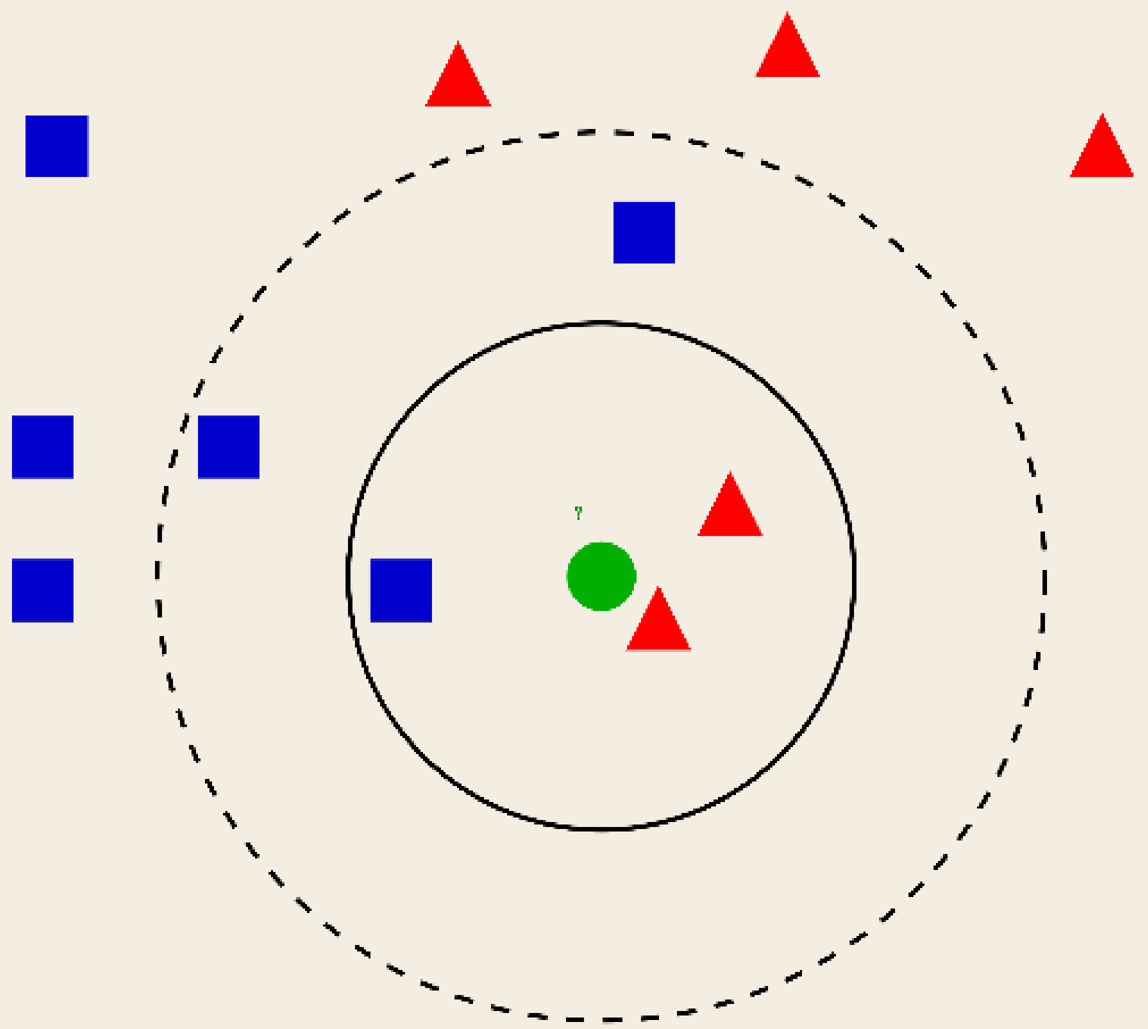
TYPES OF DATA (SOON!)



K-NEAREST NEIGHBOURS (KNN)

<u>Characteristics</u>

- Supervised: Need labelled training data
- Non-parametric: Doesn’t assume the data follows a certain distribution
- Lazy learning: algorithm does no training, it delays all computation until prediction time.

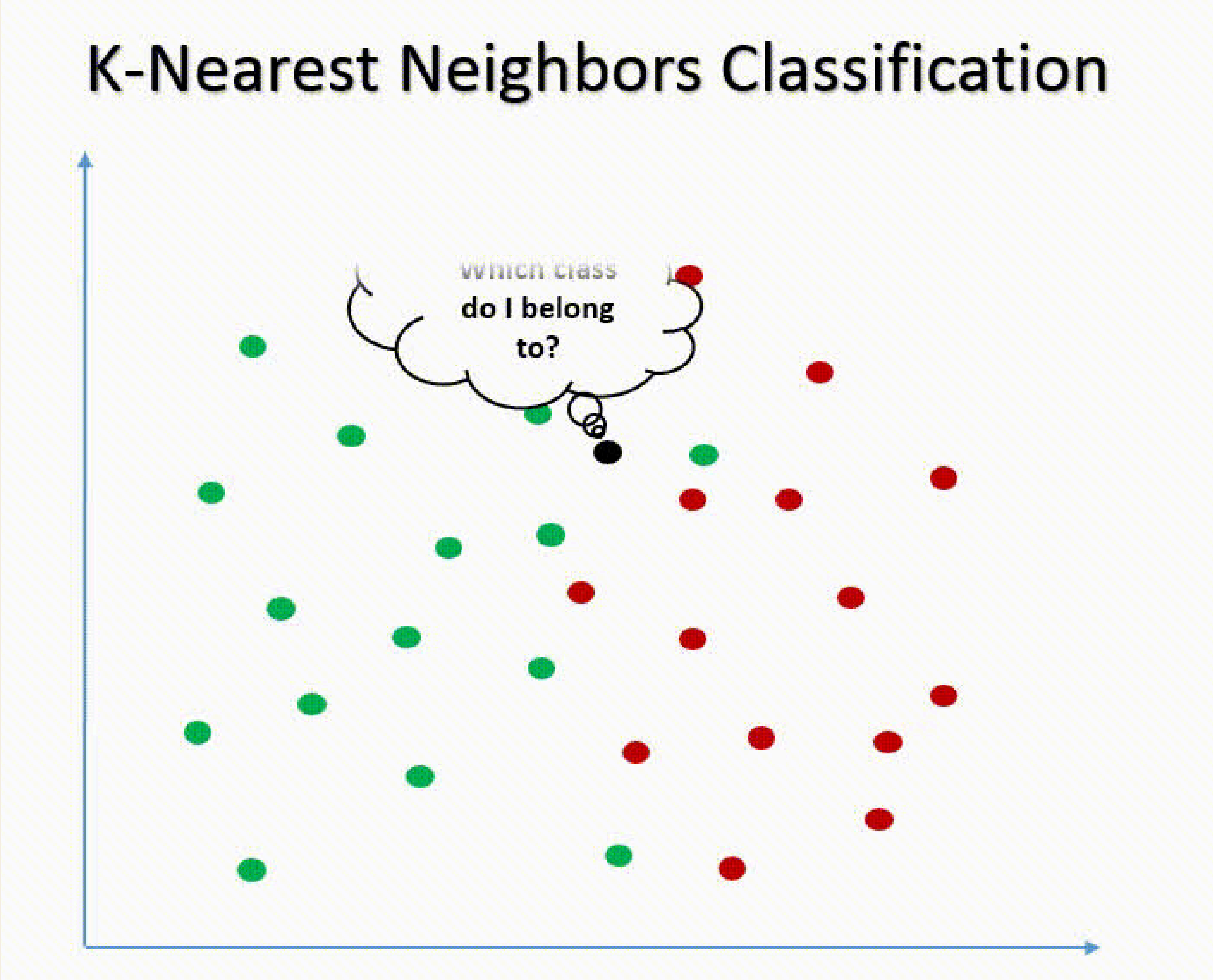


Would you assign the green circle to be blue rectangle or red triangle?

K-NEAREST NEIGHBOURS (KNN)

<u>How it works:</u>

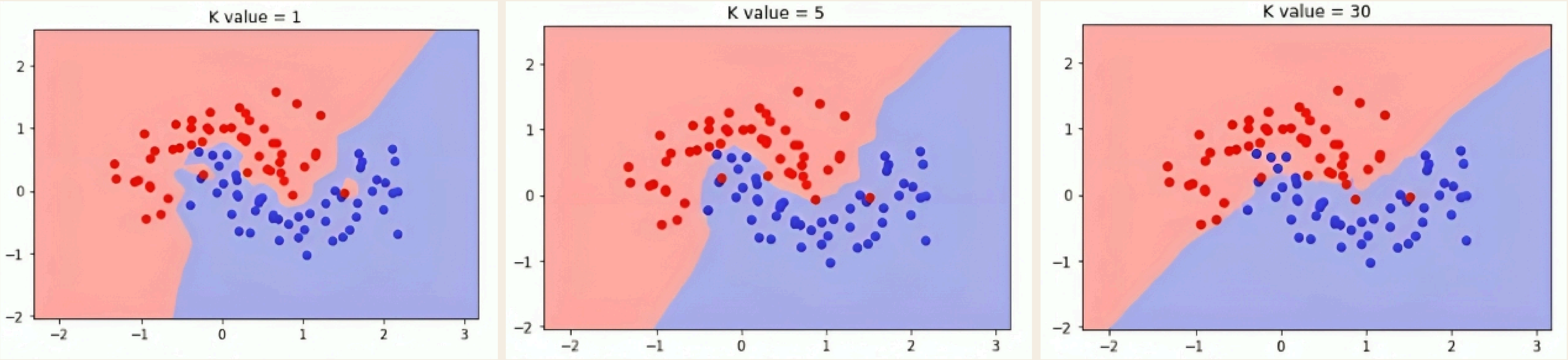
- Take a new data point.
- Measure its distance to all points in the dataset (euclidean distance)
- Find the k nearest neighbors.
- Assign the majority class as the prediction.



This is a GIF!

HOW TO CHOOSE K?

- No best guidance, so only can trial and error, but the general rule of thumb is **sqrt(N)**
- Too small k is senstive to noise (tends to overfit).
- Too large k is undefits (simplifies classification)
- **Always try to keep k as an odd integer to avoid tie between classes**



DISTANCE METRICS

For two n-dimensional data points **p** and **q**,

- Euclidean Distance: $\sqrt{\sum_{i=1}^{n}(p_i - q_i)^2}$
- Manhattan Distance: $\sum_{i=1}^{n} |p_i - q_i|$ where p_i and q_i are real values (City Block Distance)
- Minkowski Distance: $(\sum_{i=1}^{n}|p_i - q_i|^m)^{\frac{1}{m}}$ where m is the order of the norm
- Hamming Distance: $\sum_{i=1}^{n} |p_i - q_i|$ where p_i and q_i are binary values

FEATURE SCALING

- Ensure all features contribute fairly
- Prevents larger numbers (e.g., income) from overshadowing smaller ones (e.g., age).
- Helps model converge faster (Extra Readings)

#	Emp	Age	Salary
1	Emp1	44	73000
2	Emp2	27	47000
3	Emp3	30	53000
4	Emp4	38	62000
5	Emp5	40	57000
6	Emp6	35	53000
7	Emp7	48	78000

Try calculating the Euclidean distance. You'll observe that salary dominates the distance metric because of its larger magnitude, essentially drowning out the contribution from age.

WHO NEEDS FEATURE SCALING?

Typically algorithms that are sensitive to the magnitude of features.

1. **Distance-based** → KNN, K-means (W9) ✅
2. **Gradient-based** → Logistic Regression (W8), Neural Network (W10) ✅
3. **Not needed for Tree-based** → Decision Tree, Random Forest ❌

STANDARDIZATION VS NORMALIZATION

Standardisation	Normalisation
$x_{\text{stand}} = \dfrac{x - \text{mean}(x)}{\text{standard deviation }(x)}$	$x_{\text{norm}} = \dfrac{x - \min(x)}{\max(x) - \min(x)}$

We'll see more details later in Tutorial Question 4!

TUTORIAL QUESTIONS

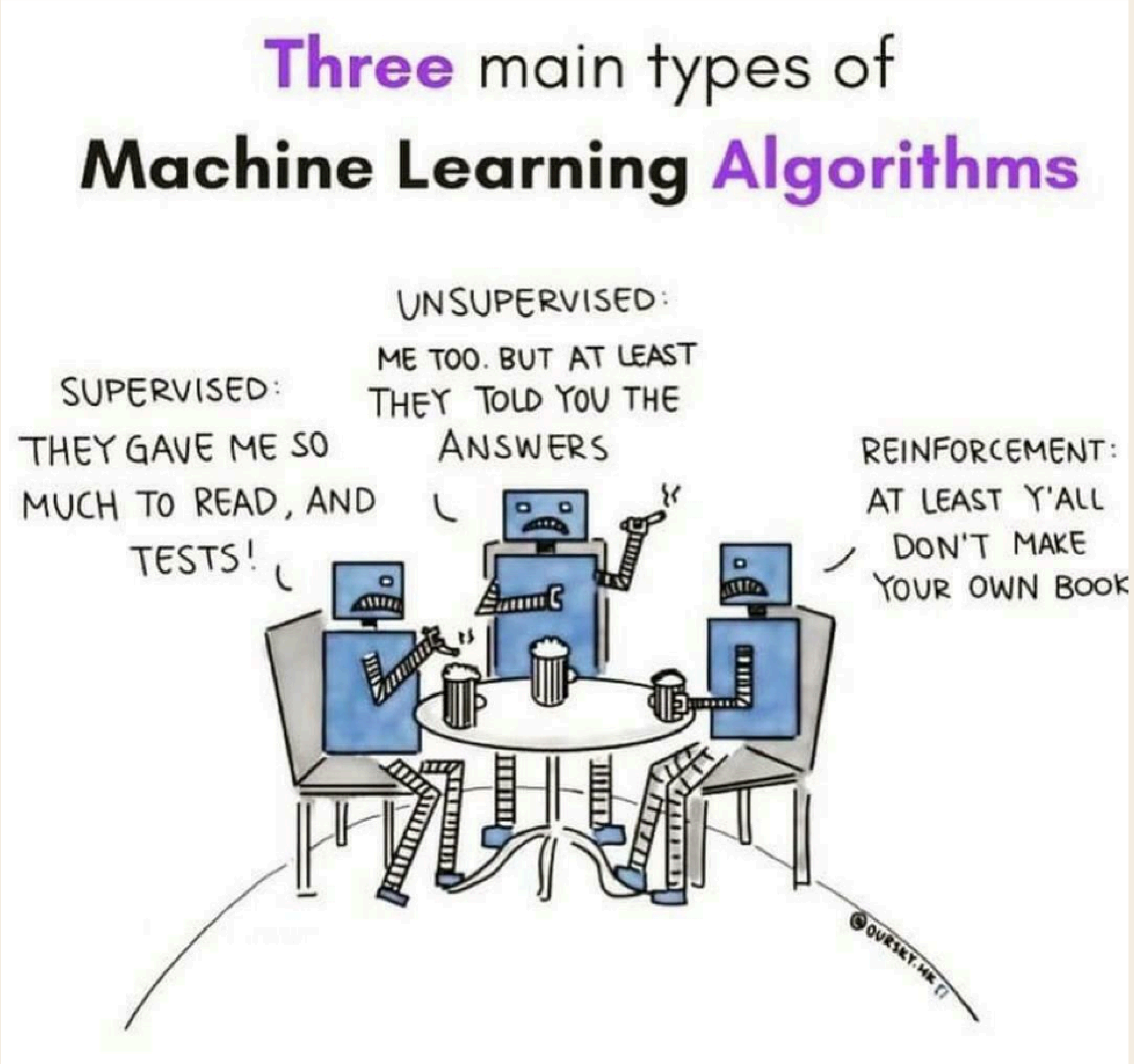
TUTORIAL QUESTION 1

Give an example Machine Learning problems for:

(i) Supervised

(ii) Unsupervised

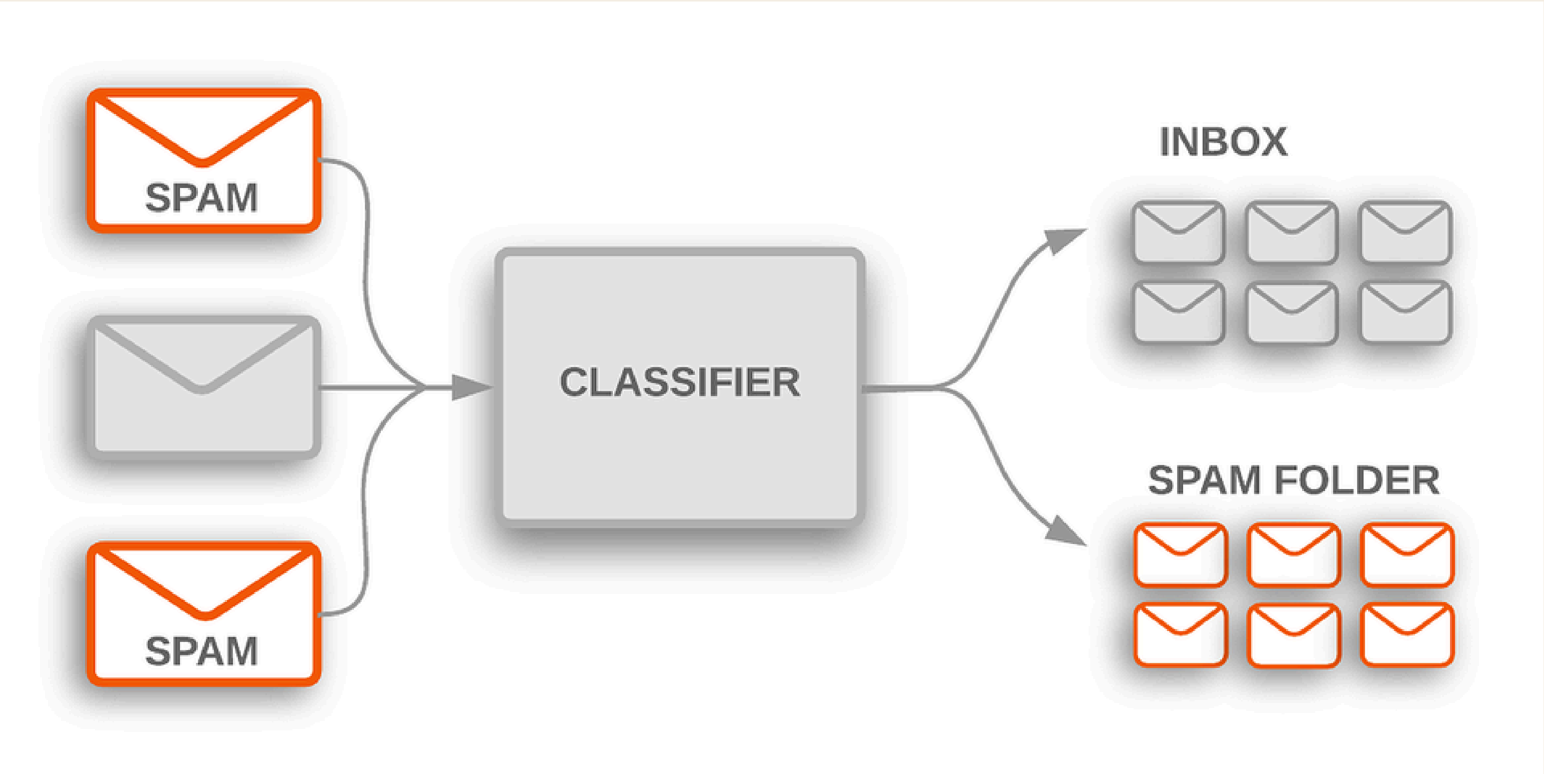
(iii) Reinforcement learning

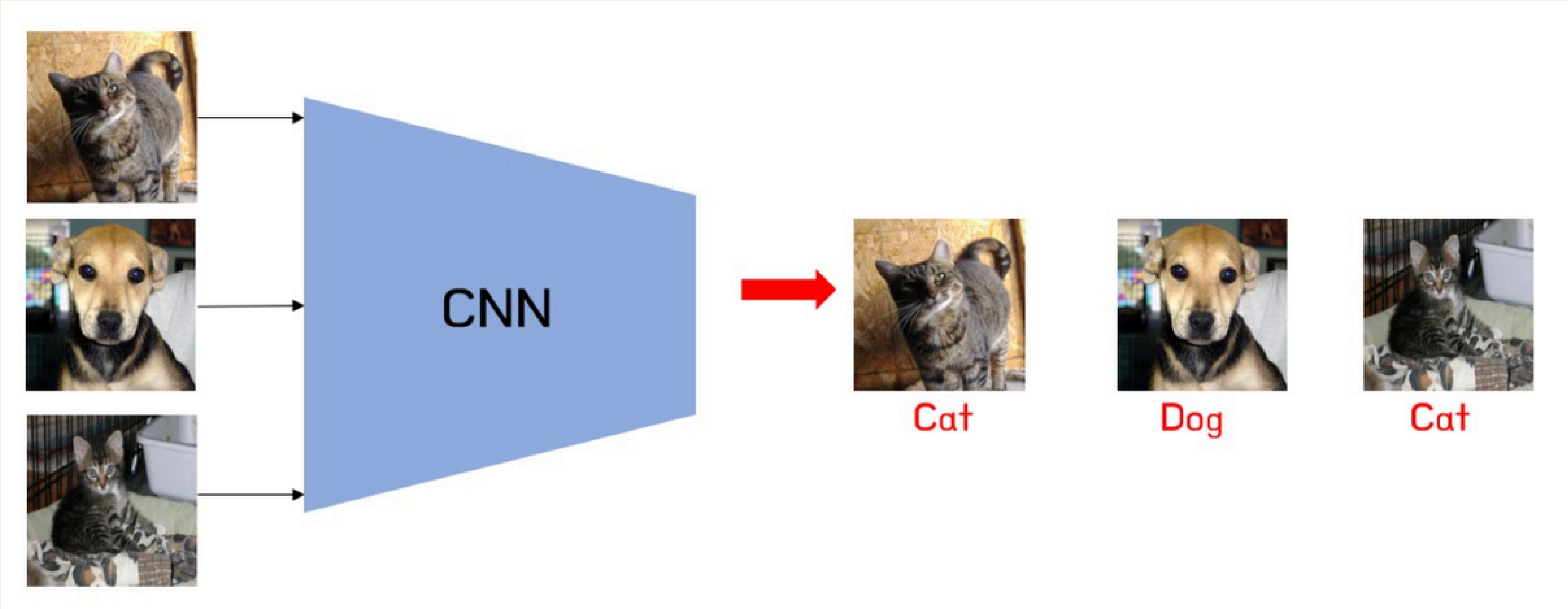


TUTORIAL QUESTION 1

Supervised Learning Examples:

1. Email Spam Detection
2. Price Prediction (Eg. Price of a resale house)
3. Image classification with Computer Vision (Eg. Objects, Faces)

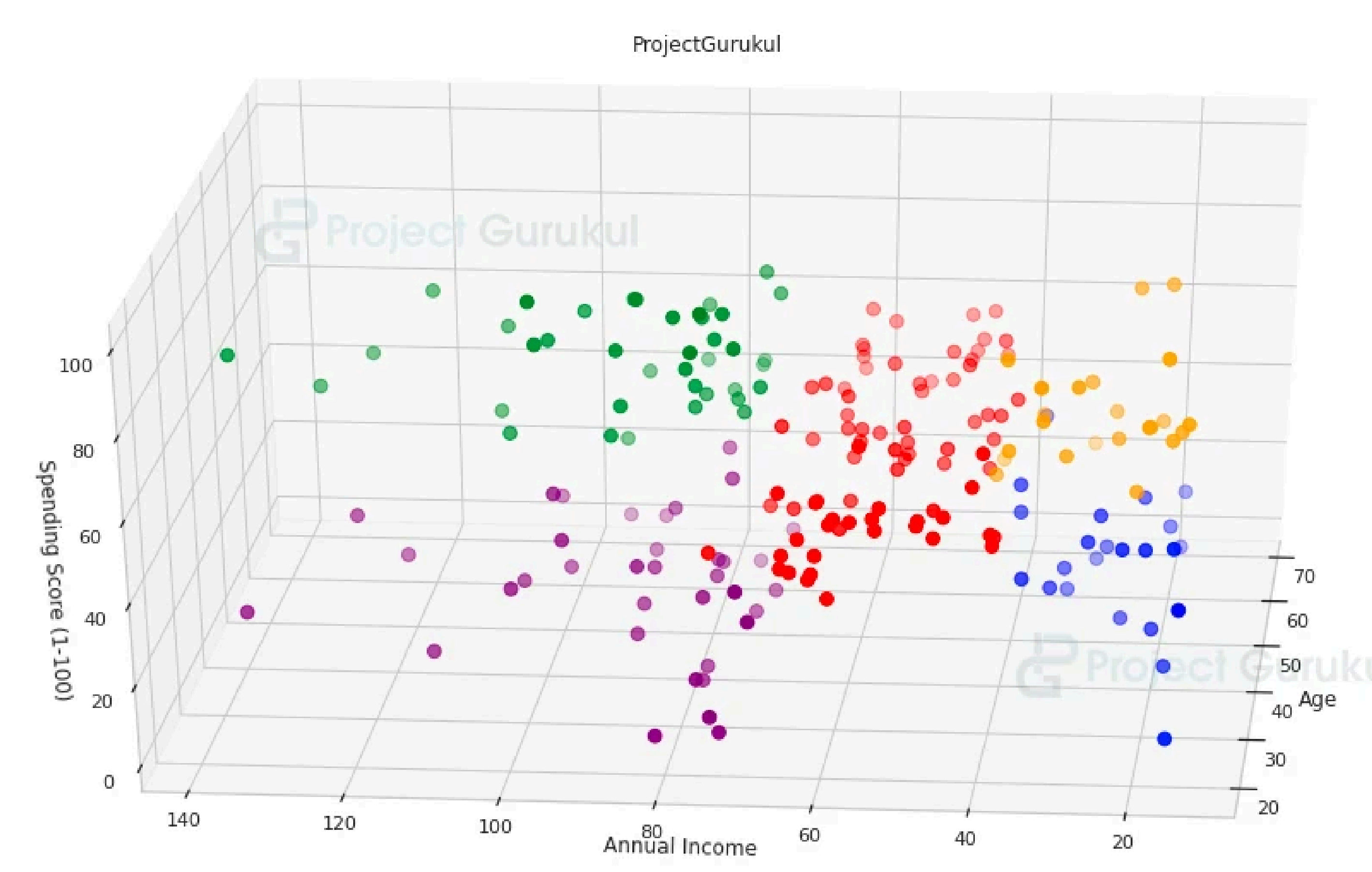


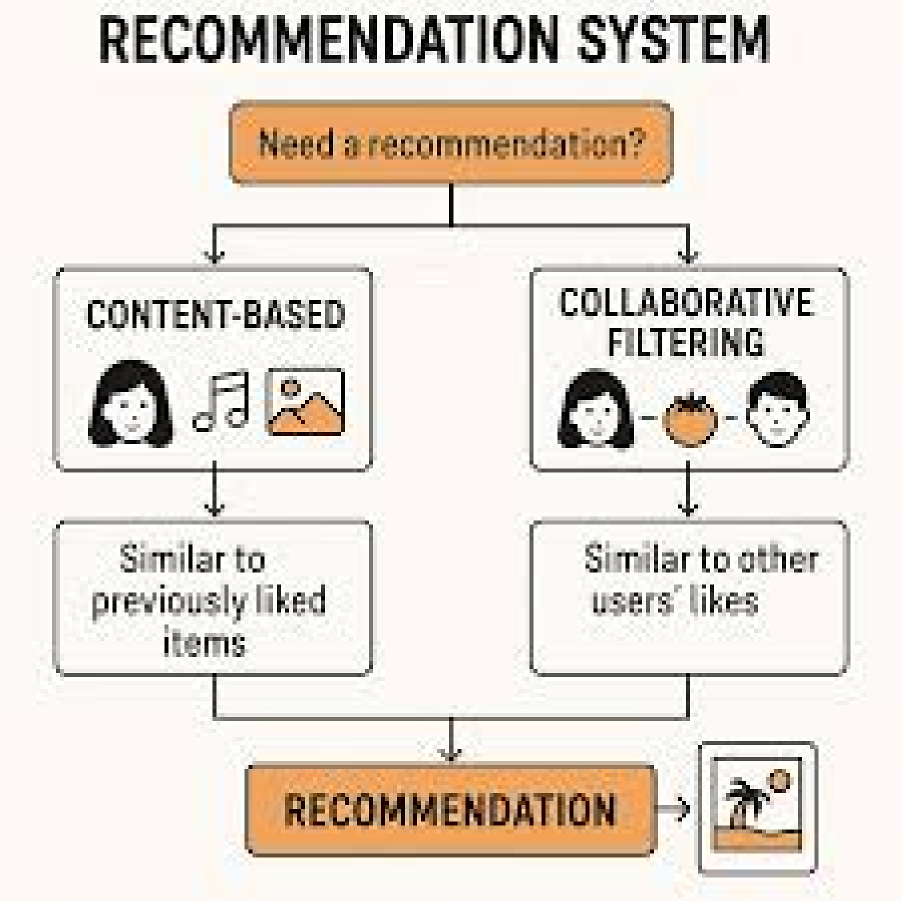


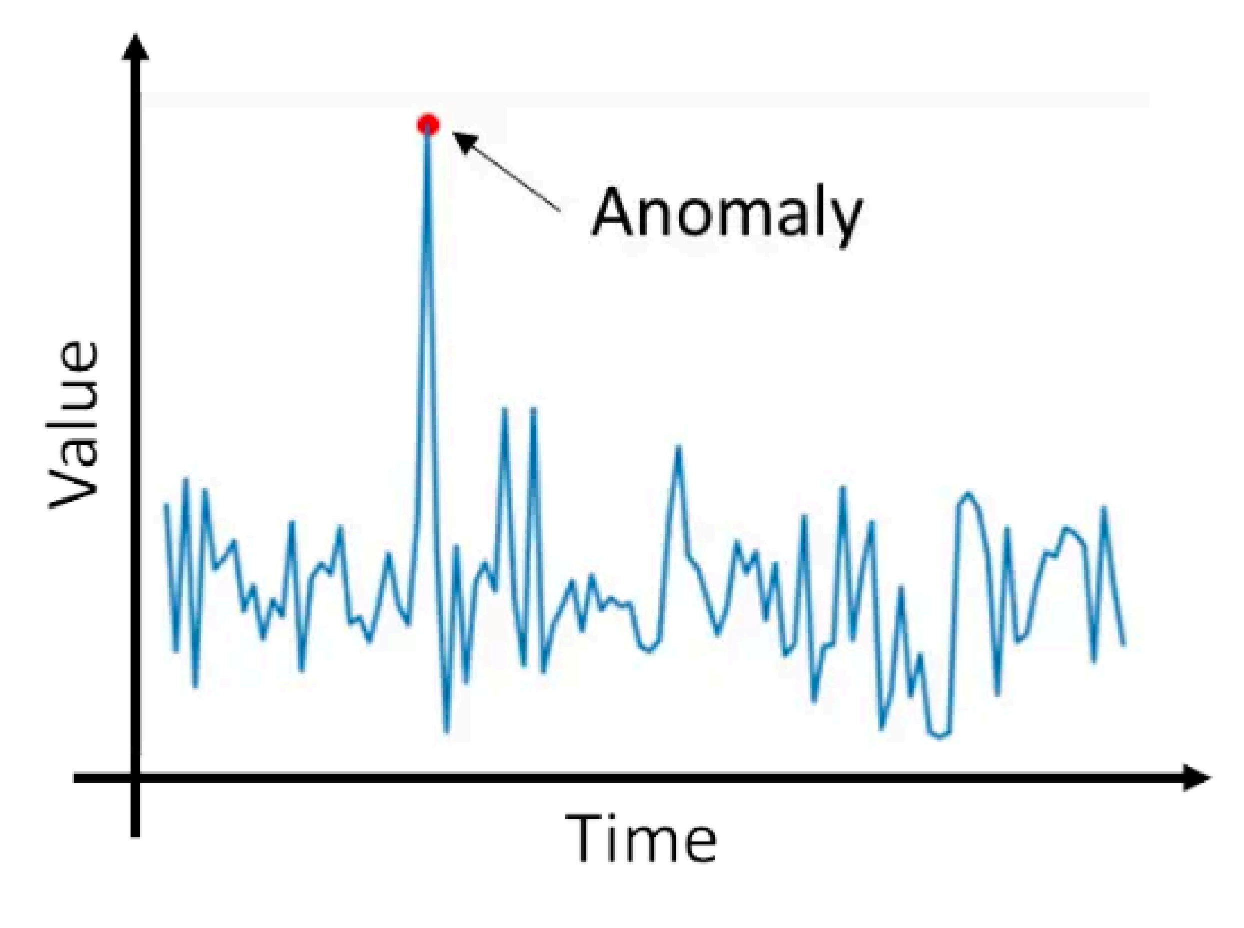
TUTORIAL QUESTION 1

Unsupervised Learning Examples:

1. Customer Segmentation (Eg. based on similar purchasing patterns)
2. Recommendation System (Eg. Spotify for songs, Amazon for products)
3. Anomaly Detection (Identify unusual patterns or outliers in data)



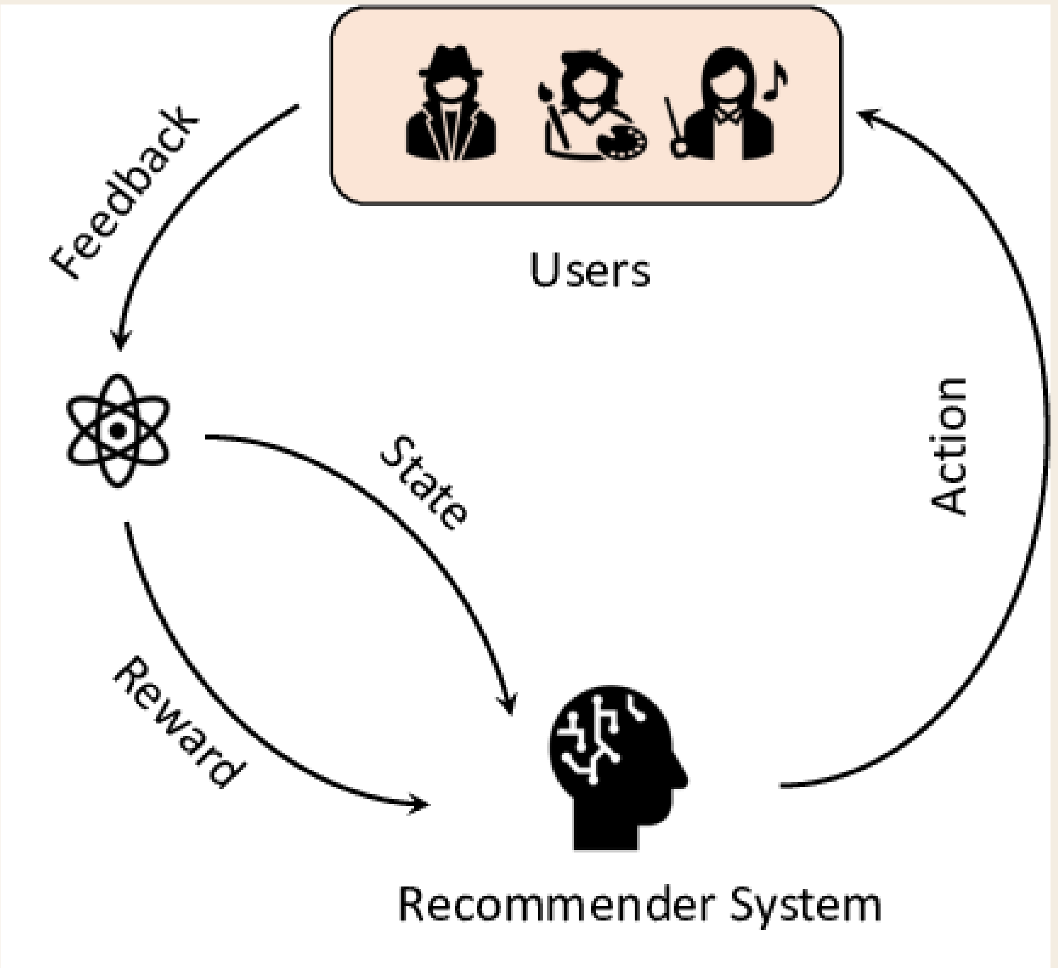




TUTORIAL QUESTION 1

Reinforcement Learning Examples:

1. AI for Games, Eg., ATARI Game AIs.
2. Autonomous Driving
3. Recommendation Systems (maximize long term user engagement)



TUTORIAL QUESTION 2

Consider the following sample dataset that contains 10 data samples, two input feature (X1 and X2), and one output feature with class 0 and 1.

	X1	X2	Y
1	4.52	3.46	0
2	4.23	2.91	0
3	2.47	4.50	0
4	4.71	5.81	0
5	3.40	4.00	0
6	8.55	5.83	1
7	6.87	4.66	1
8	10.30	3.64	1
9	8.92	4.55	1
10	9.06	1.92	1

Find out which data sample is closer to the point **(5.21, 4.32)** using Euclidean and Manhattan distance?

TUTORIAL QUESTION 2

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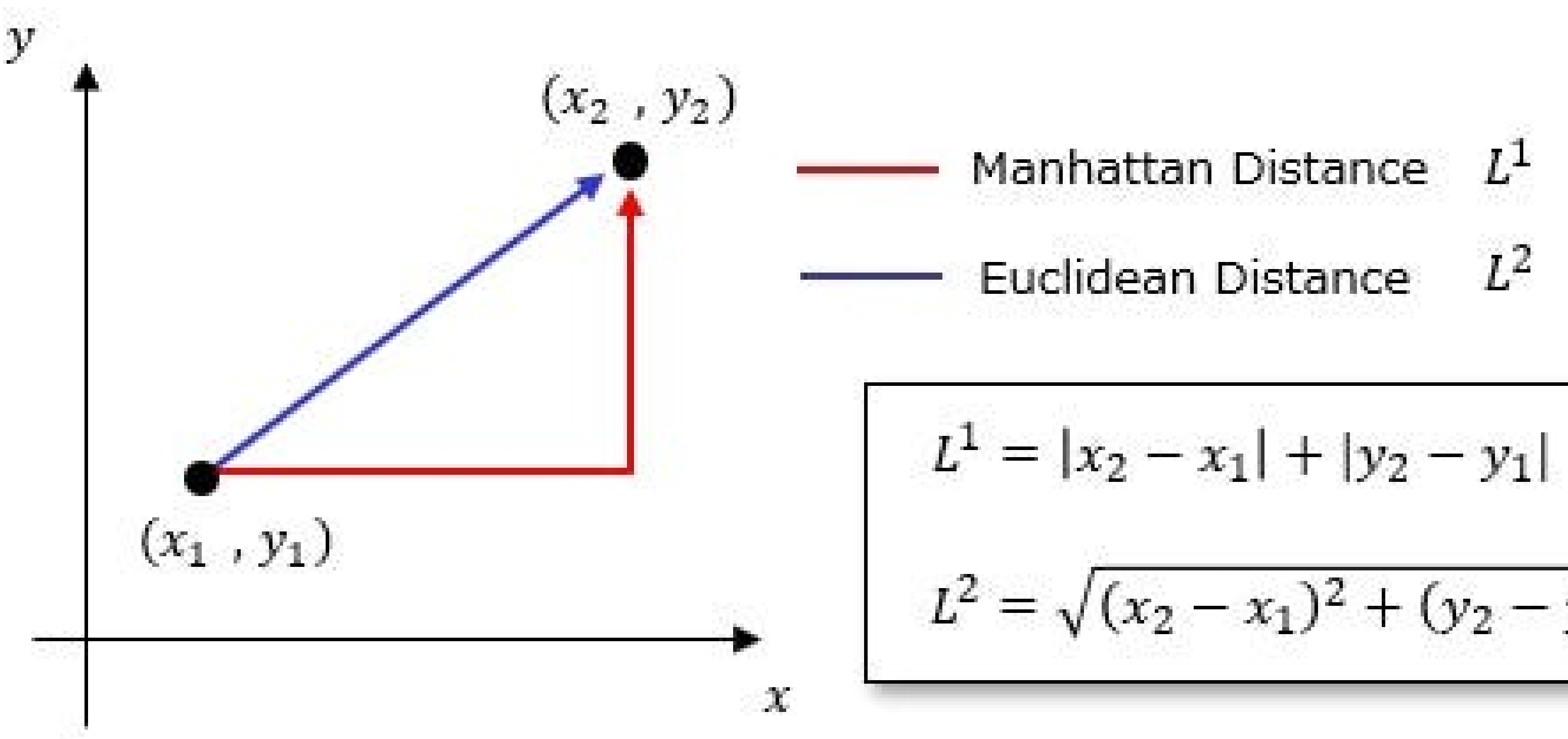

$$L^1 = |x_2 - x_1| + |y_2 - y_1|$$

$$L^2 = \sqrt{(x_2 - x_1)^2 + (y_2 - y_1)^2}$$

In exam, you can use Calculator or Excel

I personally prefer calculator.

TUTORIAL QUESTION 2

	X1	X2	Y
1	4.52	3.46	0
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10	9.06	1.92	1

Using Excel:

a. Euclidean Distance

=SQRT(($B3 - B15)^2 + ($C3 - C5)^2)

b. Manhattan Distance

=ABS($B3 - B15) + ABS($C3 - C15)

=SUM(($B3 - B15), ABS($C3 - C15))

$ locks a row and/or column in a cell reference (absolute reference)

TUTORIAL QUESTION 2

	A	B	C	D	E	F	G
1		Data					
2		X1	X2	Y		Euclidean	Manhattan
3	1	4.52	3.46	0		1.102587865	1.55
4	2	4.23	2.91	0		1.717119681	2.39
5	3	2.47	4.5	0		2.745906044	2.92
6	4	4.71	5.81	0		1.571655178	1.99
7	5	3.4	4	0		1.83806964	2.13
8	6	8.55	5.83	1		3.665474048	4.85
9	7	6.87	4.66	1		1.694461566	2
10	8	10.3	3.64	1		5.135221514	5.77
11	9	8.92	4.55	1		3.717122543	3.94
12	10	9.06	1.92	1		4.536794022	6.25
13							
14		x1	x2			min_euc_dist	min_mht_dist
15		5.21	4.32			1.102587865	1.55

Ans: Point 1 (4.52, 3.46) for both

TUTORIAL QUESTION 3

Compute the standardized feature and the normalized feature. You can also plot the original data in Q2 and two separate plots to show the standardized and normalized features.

Normalization

$$X_{new} = \frac{X - X_{min}}{X_{max} - X_{min}}$$

Standardization

$$X' = \frac{X - \text{Mean}}{\text{Standard deviation}}$$

TUTORIAL QUESTION 3

In Excel:

Standardised Data

	X1	X2	Y
1	-0.644272638	-0.547758567	0
2	-0.749061794	-0.998757386	0
3	-1.385023567	0.305039202	0
4	-0.575617674	1.379236391	0
5	-1.048975585	-0.104959725	0
6	0.811935287	1.395636348	1
7	0.204880867	0.436238858	1
8	1.444283642	-0.400158953	1
9	0.945631796	0.346039094	1
10	0.996219665	-1.810555262	1

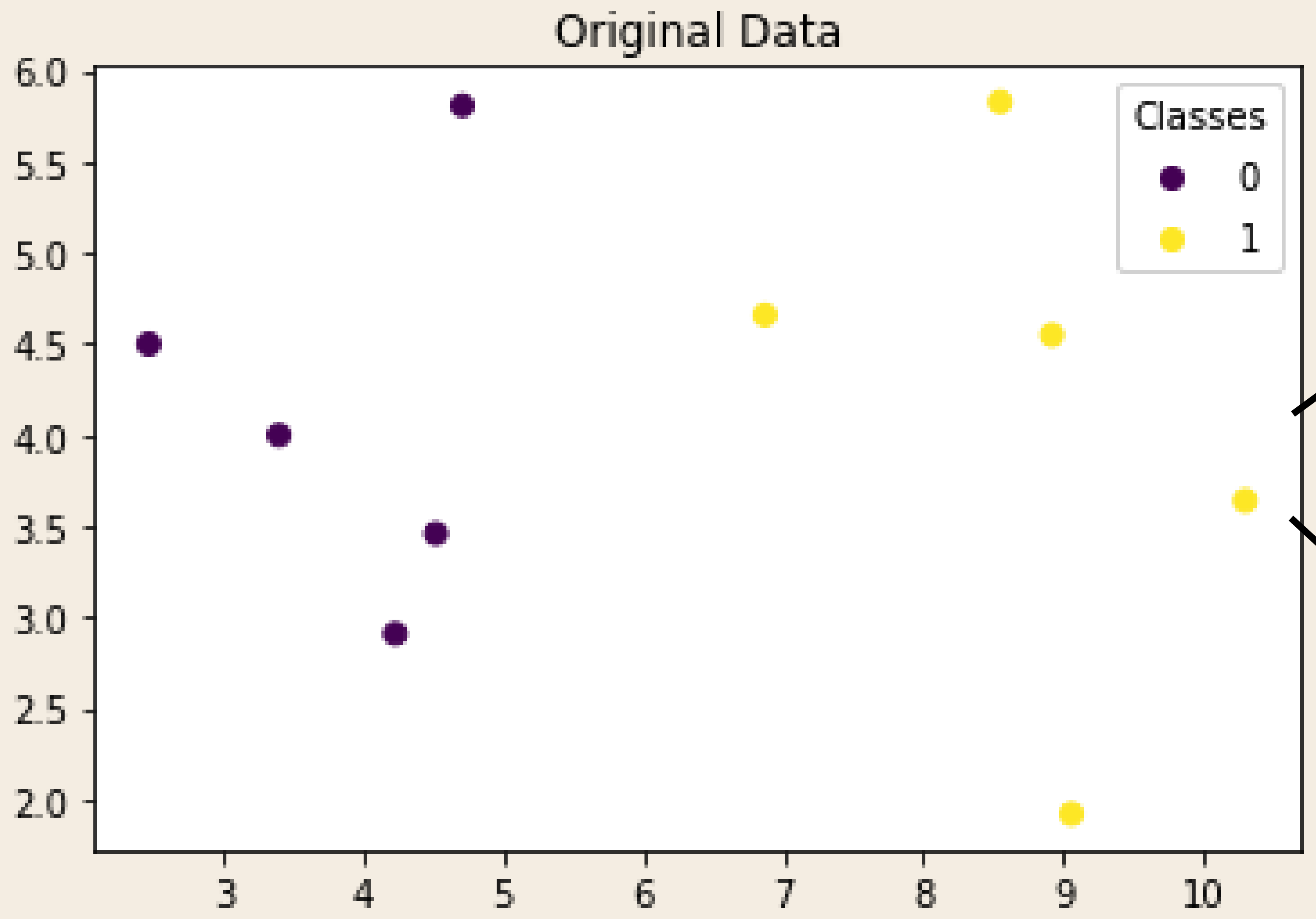
mean of X1	mean of X2
6.303	4.128
std of X1	**std of X2**
2.767461934	1.219515386

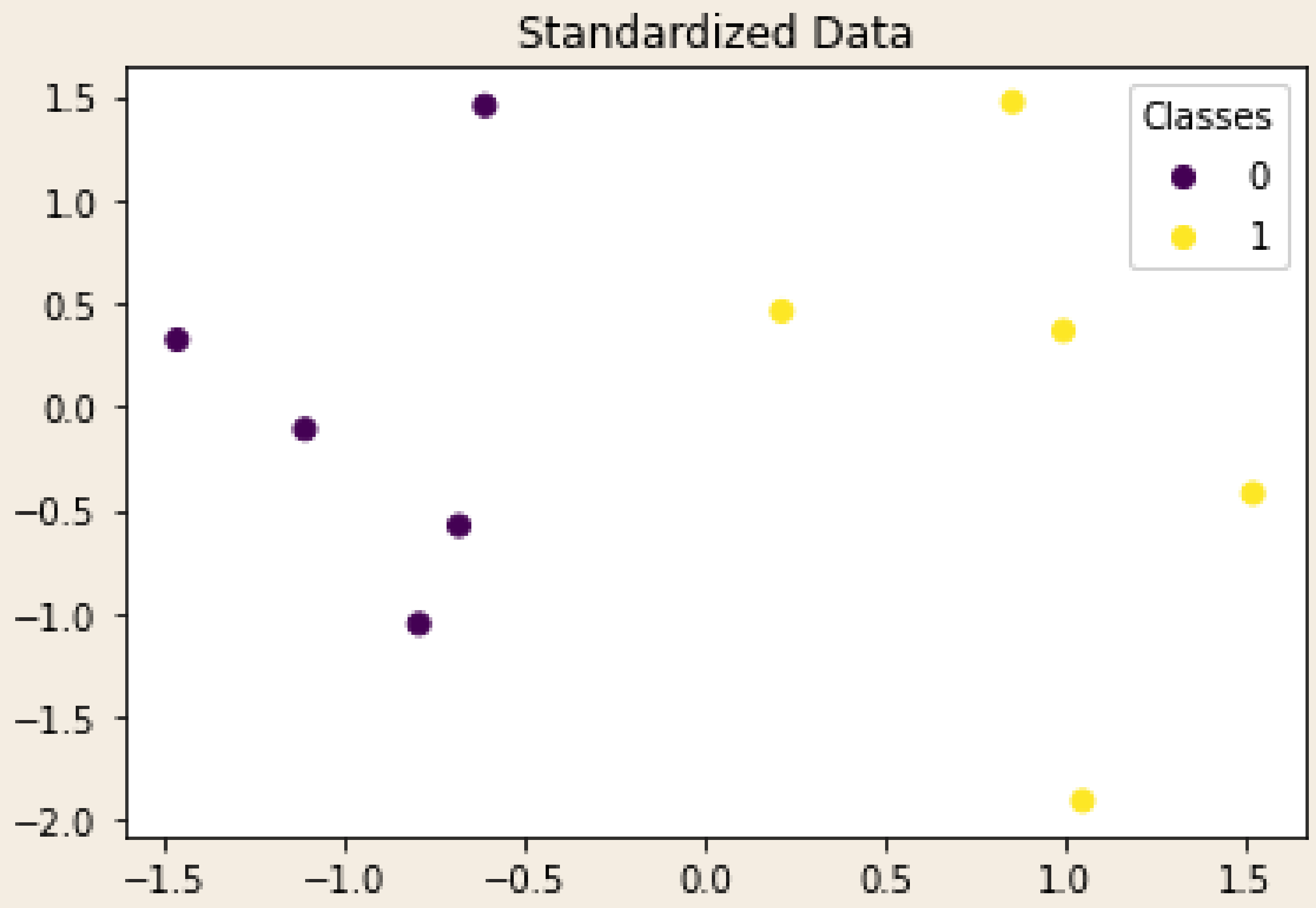
Normalized Data

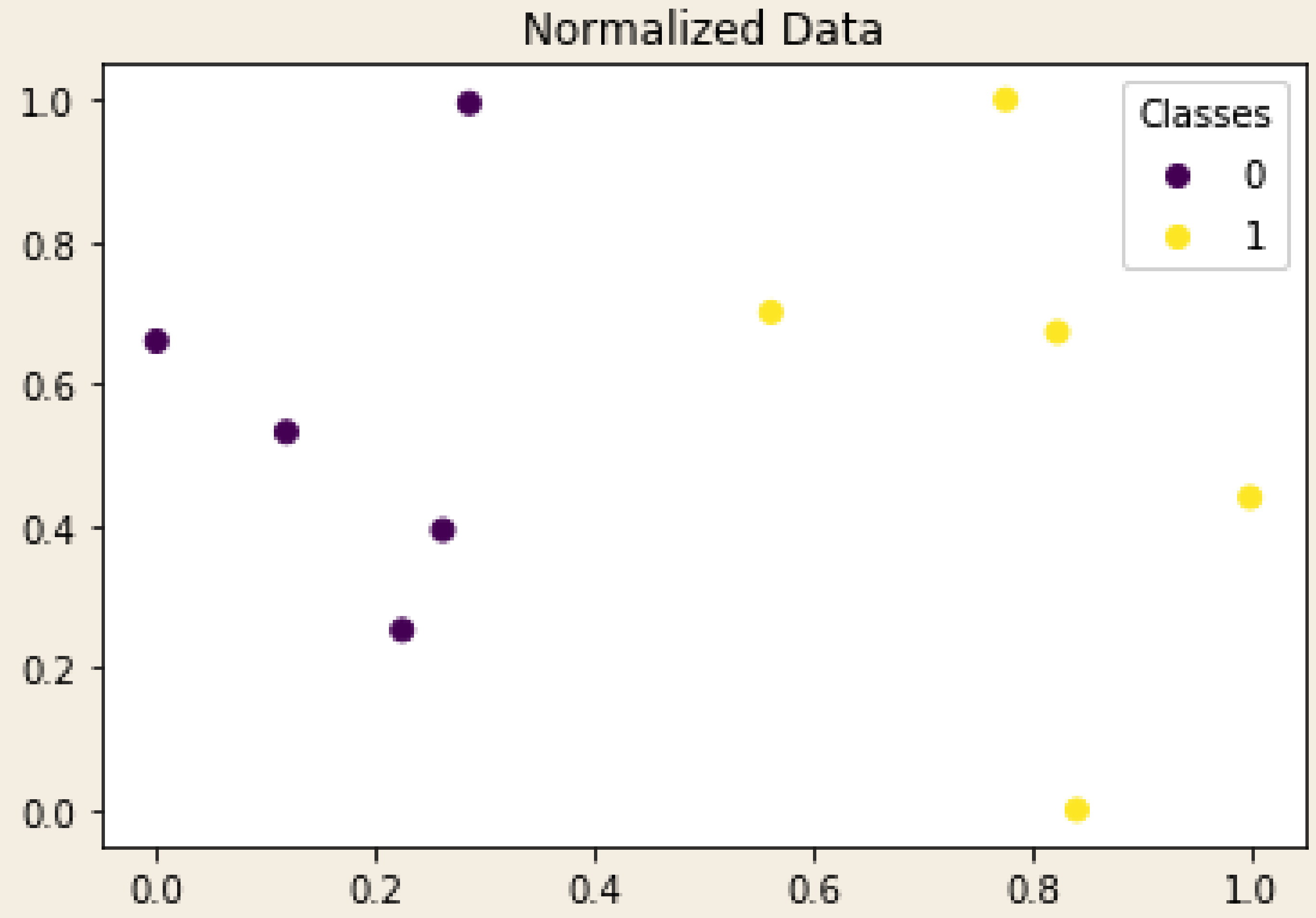
	X1	X2	Y
1	0.261813538	0.393861893	0
2	0.224776501	0.253196931	0
3	0	0.659846547	0
4	0.286079183	0.99488491	0
5	0.118773946	0.531969309	0
6	0.776500639	1	1
7	0.561941252	0.700767263	1
8	1	0.439897698	1
9	0.823754789	0.672634271	1
10	0.841634738	0	1

min of X1	min of X2
2.47	1.92
max of X1	**max of X2**
10.3	5.83

TUTORIAL QUESTION 3







colab notebook (code) will be on coursemology

TUTORIAL QUESTION 4

Discuss when you should use Standardization and when to use Normalization for feature scaling in Machine Learning tasks.

Normalization

$$X_{new} = \frac{X - X_{min}}{X_{max} - X_{min}}$$

Standardization

$$X' = \frac{\text{X} - \text{Mean}}{\text{Standard deviation}}$$

TUTORIAL QUESTION 4

Discuss when you should use Standardization and when to use Normalization for feature scaling in Machine Learning tasks.

Standardisation	Normalization
Transforms data to have mean=0 and std=1 (no fixed range)	Scales data to a fixed range 0-1 *(except new test point when its bigger than max or smaller than min)*
Less sensitive to outliers relative to Normalization (but actually still sensitive)	Sensitive to Outliers (Think: What happens is max is super large compared to other values)
Typically used with algorithm that assumes Gaussian distribution (eg. Linear Regression, Logistic Regression)	Typically used with algorithm with no makes no assumption about distribution of data (KNN)

In general, people try both and see which feature scaling methods are better.

There are also other feature scaling methods available in Scikit library.

TUTORIAL QUESTION 5

For the data given in Q2, compute the KNN to find out the class value for the new data (9.09, 4.36). You can use K=3 and Euclidean distance metric for this example.

TUTORIAL QUESTION 5

For the data given in Q2, compute the KNN to find out the class value for the new data (9.09, 4.36). You can use K=3 and Euclidean distance metric for this example.

Steps:

1. Perform normalisation on the train data. (Did in Q3)
2. **Take the metrics (e.g. *max* and *min*) calculated in (1) to normalise the new test data point (9.09, 4.36) → ELSE ITS DATA LEAKAGE**
3. Calculate Euclidean distance between Normalized Test Data point and all other normalized train data.
4. Select 3-Nearest Neighbours based on smallest Euclidean Distance
5. Take the majority vote of 3-Nearest Neighbour's class to conclude final class prediction.

TUTORIAL QUESTION 5

In excel:

Normalized Data

	X1	X2	Y
1	0.261813538	0.393861893	0
2	0.224776501	0.253196931	0
3	0	0.659846547	0
4	0.286079183	0.99488491	0
5	0.118773946	0.531969309	0
6	0.776500639	1	1
7	0.561941252	0.700767263	1
8	1	0.439897698	1
9	0.823754789	0.672634271	1
10	0.841634738	0	1

Euclidean Distance (After Normalization!)

Euclidean
0.627401597
0.7230359
0.846224003
0.671147562
0.732501706
0.382232222
0.293723174
0.240394333
0.053223088
0.624052682

3 nearest neighbours: **Final Class = 1**

new_x1	new_x2
9.09	4.36
norm_new_x1	**norm_new_x2**
0.845466156	0.624040921

IMPORTANT: When scaling data, fit the scaler on training data only, then apply it to both training and test sets.
If you calculate scaling parameters (min/max or mean/std) using test data, this is DATA LEAKAGE.

POLLEV QUESTIONS

Join by Web:

PollEv.com/kaironglee

Join by QR:

EXTRA READINGS

- KNN Imputer - for missing values in the data
- KNN and Curse of Dimensionality
- KNN Regression
- Feature scaling to make convergence faster

NEXT WEEK

Linear Regression